

RIDHAM TANEJA

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Professional Summary

AI/ML Engineer skilled in building end-to-end AI and machine learning systems for real-world applications. Experienced in workflow automation, RAG-based AI systems, model development, and deployment, with projects in hospitality automation, Fintech, and market prediction using large-scale structured data.

Skills

Languages: Python

AI & Machine Learning: LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, Multi-Agent Systems, XGBoost, LightGBM, AdaBoost, CNN, LSTM, BiLSTM

Frameworks & Libraries: FastAPI, Streamlit, Scikit-learn, TensorFlow, Keras, NumPy, Pandas, Matplotlib

Backend & Databases: REST APIs, Webhooks, Event-Driven Workflows, JSON Handling, SQLite, ChromaDB

Developer Tools: Git, GitHub, Docker, Ollama, Jupyter Notebook

Experience

AI/ML Trainee & Project Intern

October 2024 – September 2025

QMANSYS INFOSOLUTIONS (*in association with NASSCOM*)

- Participated in an industry-oriented AI/ML training and project collaboration program involving practical implementation tasks and real-world technical assignments
- Engineered a **multi-class emotion classification system** on **111K samples**, achieving **96% accuracy** using XGBoost and **94%** with LightGBM
- Improved model performance using **hyperparameter tuning**, **cross-validation**, ROC-AUC, and Precision-Recall metrics, achieving **95% validation accuracy** before deployment via **Streamlit** for real-time inference

Projects

Guestflow-AI

GitHub

- Built an **event-driven hospitality workflow automation platform** using **FastAPI**, **Streamlit**, **SQLite**, and **LangChain-compatible architecture**, simulating real-world hotel CRM and guest support operations
- Developed a **multi-agent orchestration system** that dynamically routed guest events across **Pre-Stay**, **In-Stay**, and **Post-Stay workflows** with escalation handling, VIP prioritization, and persistent guest state management
- Implemented **Retrieval-Augmented Generation (RAG)** with **local LLM inference via Ollama (Phi-3)**, enabling context-aware guest query resolution using a hotel knowledge base and AI-powered response generation
- Designed a modern **operations dashboard** with real-time workflow visualization, webhook-triggered automation, API integrations, ticket generation, and workflow analytics for hospitality support automation

UPI Sentinel

GitHub

- Designed and implemented a **CNN + BiLSTM + Attention** architecture to model transaction sequences, enabling context-aware fraud detection through deep feature engineering
- Addressed **extreme class imbalance** in **1.2M+ UPI transactions** using class weighting and threshold tuning, achieving **90% recall** while maintaining high precision
- Optimized model performance through sequence modeling and training strategies, achieving **0.85 F1-score** and **0.99 ROC-AUC** on highly imbalanced data
- Deployed a **production-oriented inference pipeline** by exposing the model via **REST API** for real-time transaction scoring, enabling practical Fintech integration

Education

Bachelor of Technology in Computer Science and Engineering(AI & ML)

Sep 2022 – Present

Raj Kumar Goel Institute of Technology, Ghaziabad, Uttar Pradesh

CGPA: 7.0/10

Class 12, CBSE Board

July 2021

Harmilap Mission School

Kanpur, Uttar Pradesh

Class 10, CBSE Board

July 2019

St. Xavier's High School

Vapi, Gujarat